

An App to Create Question Sets with Voice Input Answers

App Inventor empowers even newbies to create software applications for Android. Our regular readers have become quite familiar with App Inventor over the past few issues of *OSFY*. Noobs are advised to go through our back issues and catch up. This article is a tutorial on creating a quizzing app.



In this article let's create some popular content - a quiz with a set of questions and answers, along with an app that allows you to answer questions from a selected question set. We will also enable speech based inputs to answer questions, a novelty that could lead to other inventive applications too. Readers will require prior working knowledge of the *MIT AI 2* environment to benefit from this column.

The infrastructure

Let us examine what is needed as basic infrastructure to accomplish what we want to do. We are trying to create two apps: App1 to create data and App2 to use that data and enable some functionality. This means that data written by App1 needs to be communicated to or be available to App2. In the example featured in this article, we will use screens to simulate App1 and App2, and use TinyDB for communicating data between the apps. Refer to Figure 1. We could easily

extend this using TinyWebDB and break the screens into totally independent apps. For this article, let's restrict ourselves to using TinyDB and a single app with multiple screens, which makes it easier to illustrate concepts.

Expanding on this concept further, the elements that we see in this article are explained in more detail in Table 1.

Element	Explanation
TinyDB	TinyDB is a non-visible component that stores data for an app. It is persistent and stores data in the form of tags and values. All apps running on the same system will share their TinyDB data store.
Screen	Screen is the basic container that shows a set of GUI elements in an app. A screen can invoke other screens and retain control.
SpeechRecognizer	SpeechRecognizer is a media component that allows the developer to interact with and use voice input functionality available in Android systems.
HorizontalArrangement	This is a GUI formatting element that allows items to be shown in one line or row, arranged from left to right.
VerticalAlignment	This is a GUI formatting element that allows items to be shown one below the other in the same column, arranged from top to bottom.

Table 1: Palette elements used

Next, let's look at how the design for the TinyDB tags is arrived at. TinyDB is a simple tag <-> value store, which has a simple interface to write a tag and its value into the database. It then allows you to fetch a value based on a tag name. Now, coming to the design of tags for our 'Question and Answer' system, the idea here is to identify a set of questions with a unique text based identifier called a QuestionSet. This comprises multiple questions and their answers. Hence, we need to know how many questions are present in a question set. We also need a way to identify all the QuestionSets already created in the system. So, we use the tags as explained in Table 2 for storing and retrieving the values needed on our app screens.